

Kick-Off Meeting Result Protocol

Understanding the compensation policy impact on local resident's tolerance towards wild boars in Giant Panda National Park, China

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Munich, March 03rd, 2026

Start Time: 14:03

End Time: 14:57



Submitted in partial fulfillment of the requirements for the consulting project module.

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1 Project Background

Mengxi Rou firstly describes the big picture of her research, explain why she focuses her research on wild boars in China, then elaborates on the key word of the project which is Compensation Policy and also mentions Wildlife Tolerance Model.

Topic: Can compensation policy increase tolerance toward wild boars? Understanding the compensation policy impact on local resident's tolerance towards wild boars in Giant Panda National Park, China

This project investigates whether participation in wildlife compensation policies influences local residents' tolerance toward wild boars in the Giant Panda National Park in China. Human–wildlife conflict is a major issue in conservation, and financial compensation is a key policy tool used to reduce economic losses caused by wildlife. The study is based on the Wildlife Tolerance Model (WTM), which distinguishes between:

- Inner Factors (e.g., values, attitudes, social norms), and
- Outer Factors (e.g., costs, benefits, and policy-related variables)

Due to measurement limitations, the analysis focuses only on the outer model. This represents an important limitation, as previous research suggests that inner factors may be stronger predictors of tolerance. Preliminary results indicate that participation in compensation schemes is associated with lower tolerance toward wild boars, which contradicts common policy expectations and motivates further analysis.

2 Data

The data used in this study originates from a cross-sectional household survey conducted in two counties within the Giant Panda National Park:

- Baoxing County ($n = 160$ after filtering, August 2024)
- Qingchuan County ($n = 465$ after filtering, March–April 2025)

The total sample size is $n = 625$, including:

- 277 households with compensation
- 348 households without compensation

The two counties are largely comparable in terms of sociodemographic characteristics; therefore, the datasets were combined for the analysis.

Key Variables

Tolerance:

- Willingness to allow killing
- Support for population control
- Perceived safety threat
- Ability to absorb crop loss

Compensation Participation: Binary indicator (Yes/No)

Conflict Experience:

- Wild boar specific (frequency, duration, population trend)
- General (species number, severity, duration)

Tangible Costs:

- Damaged area
- Perceived economic loss

Intangible Costs:

- General attitude toward wild boars
- Descriptive word associations

Intangible Benefits:

- Perceived ecological value

Prevention Effectiveness:

- Average perceived effectiveness of mitigation measures (excluding compensation)

Compensation Anticipation:

- Lowest accepted percentage
- Expected percentage
- Annual insurance payment

Policy Support and Dependency:

- Reported institution
- Knowledge source

Sociodemographics:

- Age
- Gender
- Education
- Income
- Household assets

The dataset was shared after the initial project meeting. No data protection or intellectual property restrictions apply. All analyses will be conducted using the statistical software R.

3 Research Questions and Analytical Approach

RQ1. Does compensation for wild boar damage have a direct effect on tolerance toward wild boar, or is its influence mediated by other social and psychological factors?

Existing model results suggest that compensation participation has a significant negative direct effect on tolerance ($\beta = -0.115$, $t = -3.309$). While this direct relationship appears robust, the potential indirect effects have not yet been formally tested.

In particular, two mediation pathways will be examined in the next step. First, the pathway *Compensation* $\rightarrow$ *Policy Dependency* $\rightarrow$ *Tolerance* appears to be an important driver within the compensation group, as policy dependency seems to be a significant factor. Second, the pathway *Conflict Experience* $\rightarrow$ *Intangible Costs* $\rightarrow$ *Tolerance* will be analyzed, since the direct effect of conflict experience on tolerance is non-significant, but an indirect effect via intangible costs is suggested by the current results.

These mediation effects will be tested using bootstrapped confidence intervals to assess their statistical significance.

RQ2. What roles do tangible costs, intangible costs, and intangible benefits play, and how do they interact with compensation experience?

The existing models already show clearly that intangible costs and benefits are the main drivers of tolerance, while tangible costs are not significant.

What remains to be tested is whether compensation experience changes the strength of the intangible cost effect. The estimated β values suggest such a moderation effect (-0.350 without compensation vs. -0.178 with compensation), which will be formally tested by including an interaction term in the model.

RQ3. How robust are these relationships across model specifications and groups?

To assess the robustness of the findings, several steps will be undertaken. First, a comparison table of key path coefficients across all model versions will be created to identify which results are stable and which depend on model specification.

Second, sensitivity analyses will be conducted. For example, Model 0 (RQ1) will be re-estimated without sociodemographic variables and without Conflict Experience to examine whether the main compensation effect remains consistent.

Third, a formal multi-group analysis (MGA) will be performed to test whether differences between the compensation and non-compensation groups are statistically significant, rather than relying on visual comparisons of sub-group models.

Finally, additional model evaluation metrics, including predictive relevance (Q^2), effect size (f^2), and the Standardised Root Mean Square Residual (SRMR), will be reported to provide a more comprehensive assessment of model quality.

4 Methodological Challenges

1a. How can the structural model be improved?

To improve the structural model, a moderation term (*Compensation* $\times$ *Intangible Cost* $\rightarrow$ *Tolerance*) will be included to test whether compensation alters the strength of the relationship between intangible costs and tolerance. In addition, the mediation pathways identified in Research Question 1 (RQ1) will be formally tested. Sociodemographic variables will be treated as control variables, as they are non-significant across all current model specifications.

1b. Should constructs be merged, re-specified, or modelled formatively vs. reflectively?

Based on the current results, some indicators (e.g., Tol3 and CA1/CA2) could be removed due to consistent non-significance across all models. Furthermore, it will be explored whether the two Conflict Experience constructs should be merged, as *Conflict Experience – All* drops out in sub-group models but remains in the full model.

2a. Construct validity and reliability

All standard criteria will be assessed, including Average Variance Extracted (AVE), composite reliability, and the Heterotrait–Monotrait ratio (HTMT), across all model versions and reported systematically. The HTMT check is currently missing and will be incorporated.

2b. Indicator selection and redundancy

Based on the existing results, some indicators (e.g., CA1, CA2, and IC2_3) could potentially be removed, as they are non-significant across all model versions. Removing these indicators would simplify the model without a substantial loss of information.

3a. Sensitivity checks

Path coefficients will be compared across all model versions in a summary table. In addition, Model 0 will be re-estimated without sociodemographic variables and without *Conflict Experience – All* to assess whether the main compensation effect remains stable. Furthermore, additional evaluation metrics, including predictive relevance (Q^2), effect size (f^2), and the Standardised Root Mean Square Residual (SRMR), will be reported.

3b. Multi-group analysis vs. simplified models

A formal multi-group analysis (MGA) will be conducted to test whether differences between compensation and non-compensation groups are statistically significant. Simplified model versions will be retained as robustness checks and reported in the supplementary material.

5 Project Timeline

The project is expected to be completed within eight weeks and is structured into five main phases:

- **Phase 1: Project Setup, Data Understanding, and Background Review**
The project is expected to be completed within eight weeks and is structured into five main phases:
- **Phase 2: Measurement Model Evaluation and Model Development**
The measurement model will be specified by defining latent variables and their corresponding indicators. Reliability and validity will be assessed using standard criteria, including outer loadings, variance inflation factors (VIF), and Average Variance Extracted (AVE). Based on these results, the model will be refined by removing non-significant or redundant indicators.
- **Phase 3: Structural Analysis and Robustness Checks**
The structural model will be estimated using PLS-SEM to test the hypothesised relationships between compensation policy and tolerance. Robustness checks will be conducted, including sensitivity analyses and a formal multi-group analysis (MGA) to examine differences between compensation and non-compensation groups.
- **Phase 4: Interpretation and Theoretical Refinement**
The results will be interpreted in the context of the Wildlife Tolerance Model. Based on the findings, the theoretical framework will be refined, and policy implications will be derived.
- **Phase 5: Final Report and Presentation**
The final phase includes the preparation of the written thesis, the presentation of results, and the documentation of all analyses and findings.

Mengxi Rou is planning to present her findings at the beginning of June 2026. Therefore, our consulting project should be completed beforehand, ideally by the end of May 2026. The exact deadline will be agreed upon at a later stage. Communication with the project partner will take place via email, and no special technical infrastructure is required.